Phase changes

antocciphysics | TEACHER KEY | CPA

Learning target: Account for energy transfer during a phase change without a temperature rise.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 1; 3 course days later.*
Temperature change describes a thermal-state change (°C or K); heat is transferred energy (J). They are not the same quantity.

02. Explain a worked example (worked)
Rising temperature before melting, flat during phase change; plateau length corresponds to 200 J for this amount.

03. Look and explain (model)
Flat segment is phase change; energy changes intermolecular arrangement rather than increasing temperature.

04. Build the solution (practice)
600 J.

05. Use the idea (reason)
Need latent heat mL; zero temperature change does not imply zero energy transfer.

06. Choose a method (mixed) *Review: Unit 4, Day 14; 12 course days later.*
24 N toward the natural length.

07. Exit check (exit)
300 J.

08. STRETCH: Apply it another way (stretch)
100 and 300 J; energy changes the material's phase/internal arrangement.

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01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 1; 3 course days later.*
No. Internal energy depends on amount of material and microscopic/material/phase structure as well as temperature.

02. Explain a worked example (worked)
Rising temperature before melting, flat during phase change; plateau length corresponds to 200 J for this amount.

03. Represent and interpret (model)
Flat segment is phase change; energy changes intermolecular arrangement rather than increasing temperature.

04. Solve and justify (practice)
33, 400 J melting; 8400 J warming; 41, 800 J total.

05. Analyze the reasoning (reason)
Need latent heat mL; zero temperature change does not imply zero energy transfer.

06. Choose a method (mixed) *Review: Unit 4, Day 14; 12 course days later.*
F=-kx=+12 N, right.

07. Exit check (exit)
L 225, 000 J/kg; Q=-45, 000 J.

08. STRETCH: Apply it another way (stretch)
Melting 66800 J plus warming 8400 J=75200 J; separate the two processes.